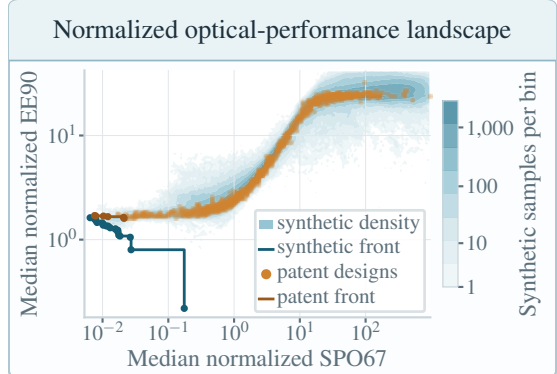
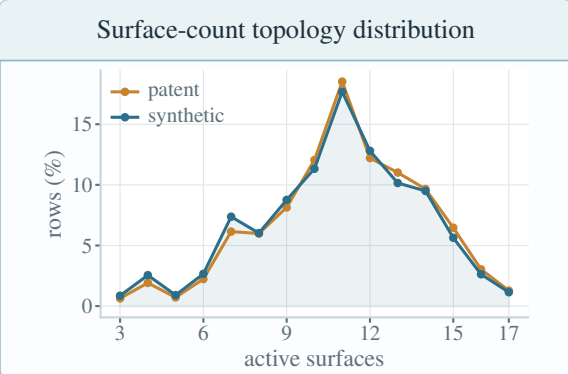
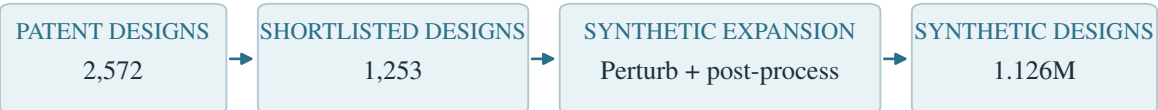


A Traceable optical data expansion



2,572 curated patent designs

1.126M synthetic designs

B Optics-aware representation and generation

What changes in the representation

Lens spacing → overall length + relative gaps

Curvature → scaled to lens length

Semi-aperture → scaled to entrance pupil

Material state → glass or air at each slot

Stop position → relative location in lens

Why it helps

✓ Avoids extreme radius values

✓ Compares geometry across lens sizes

✓ learns air gaps and stop location

↓ Material issues 15.93% → 1.55%

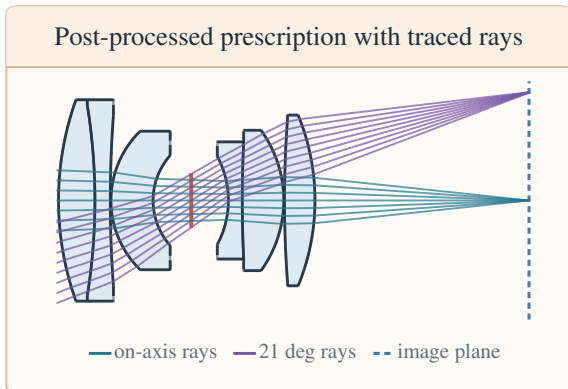
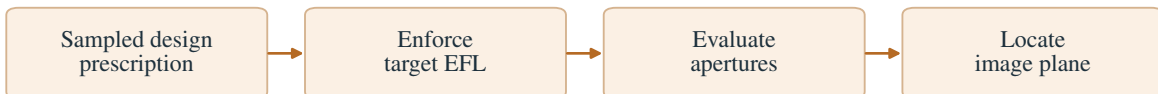
↓ Geometry issues 10.40% → 0.42%

↓ Ray-tracing errors 34.53% → 14.73%

49.51% → 83.72%

Feasibility improvement through tuned representation

C Physics-based post-processing



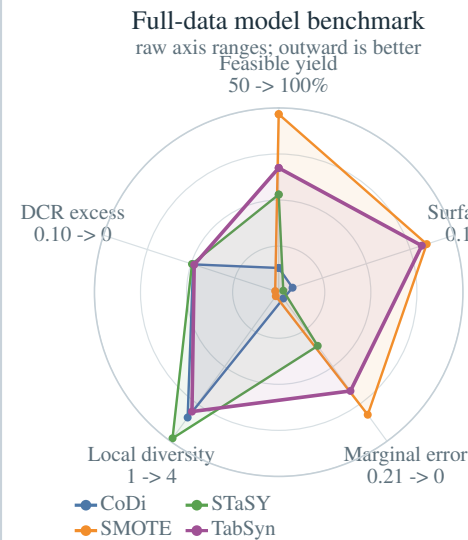
Performance improvements with post-processing

	SPO67	SPO100	EE50	EE90	Nominal
0	92%	86%	43%	34%	85%
1	88%	80%	37%	30%	78%
2	83%	74%	32%	25%	69%
3	76%	67%	27%	21%	60%
4	61%	52%	21%	17%	45%

67.29% → 83.72%

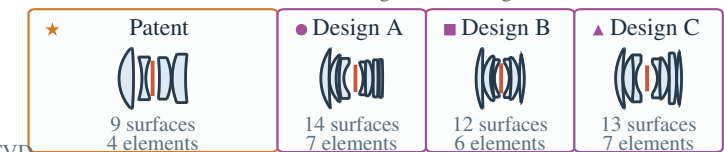
Feasibility improvement through post-processing

D Model benchmark and condition-matched designs

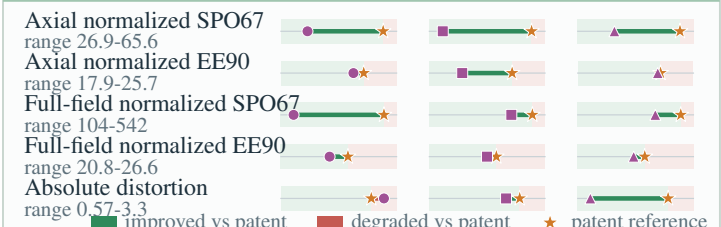


Patent-matched high-performing TabSyn designs

Reference F/# 1.79, EPD 55.7; generated designs within 10%



Patent-relative optical performance | log scales | lower is better



83.72% feasible

TabSyn: strongest learned-model balance